

AUTOCAR RECRUITMENT GAME 2023

1. Introduction

- 1.1 This autocar competition was designed to guide the participants in the basics of mechanisms, electronics and programming learning.
- 1.2 At the end of this competition, participants will be able to design, create their own personal autocar.
- 1.3 **Active** and talented participants may be scouted and may have the opportunity to join the UTM Robocon Team 2023/2024.

2. Game overview

- 2.1 Participants are required to fabricate and program an autocar to complete the tasks which had stated on the game rules.

3. Safety guidelines

- 3.1 **SAFETY FIRST.**
- 3.2 Participants must pay utmost attention during the whole Autocar Mentorship to prevent any accidents from happening.
- 3.3 Please inform any mentors while facing malfunctions.
- 3.4 Participants must ask for permission from any mentor before using any tools and materials in the lab.
- 3.4 All tools in the lab must be handled with care and used under any mentor's vision.

4. Autocar specifications

- 4.1 Before the game starts, the autocar's dimensions **must not exceed 20cm x 20cm x 20cm.**
- 4.2 There is **no restriction** on the autocar's dimensions **during** the match of the game.
- 4.3 Power supply for autocar should not exceed 12V. AC power source is prohibited.
- 4.4 Mass for an autocar is unlimited.
- 4.5 Circuits boards used must be designed by the participants and no other commercial boards or MCUs except blue pill, are

allowed.

4.6 Autocar which **not follow** any specifications above are **not allowed** to take part in both games.

5. Game Flow

- There are **two stages** in the Autocar Recruitment Challenge 2023.

Stage 1: “All Lines Lead to Robocon Lab”

Main Task: Participants are required to program an **autonomous line follow** autocar to **get** as many **points** as possible, and **reach Finish Zone** within the time given.

5.1. Before the game starts:

- a) **2 contestants** (red and blue respectively) will be given **1 minute** to **set up** their autocars in their respective start zones of the game fields. (refer to appendix for game field specifications)
- b) During set up time, contestants may arrange their autocar position and test their autocar performance in the game field.
- c) Contestants are **not allowed** to **touch** their autocar **after the set up phase has ended**.
- d) Both contestants' autocar should be inside the Start Zone, including its space above, after the set up phase has ended.
- e) Autocars are allowed to stay “switch on” while waiting for the game to start.

5.2. During the game:

- a) Game length: **3 minutes**.
- b) Contestants may press the push button or switch on the autocar to start competing.
- c) After the autocar completely moves out from the Start Zone, contestants are **not allowed to touch** the autocar **EXCEPT for retry**. (please refer to “RETRY” section)
- d) Autocar should **follow the lines autonomously**. A violation might be given by judges if the autocar failed to do so. (please refer to “VIOLATION” section)

- e) Red or blue dots on the game fields are **checkpoints**. Passing checkpoints may grant the contestant some points. (please refer to "POINTS" section)
- f) "**Arrow**" on the gamefield means **one-way**. Autocar must follow the directions ,where the direction of the arrow is pointed on that track.
- g) There is a ping pong ball located in the red/blue ping pong ball zone. Contestants **MAY** request picking it manually from the judges.
- h) After picking the ping pong ball, the autocar should stay inside the marble area before changing back to autonomous mode.
- i) If the ping pong ball has been picked, **gate 1** of your **own** gamefield will **open**, while **gate 2** at your **opponent's game field** will **close**.
- j) Stage 1 game will end if:
 - 3 minutes of game time has ended.
 - **One of the contestants** (red or blue) autocar has **reached** the **finish zone**
 - **Both** autocars are **unable to move** for **10 seconds**.

5.3. After the game

- a) Total points earned by each contestant will be recorded.
- b) Both contestants may switch off their autocars and leave the game field.

5.4. Points

Checkpoints	Points	Checkpoints	Points
A	1	F	3
B	1	G	X2 of your total mark after game ended (must release the marble at this checkpoint to earn it)
C	2	H	2
D	2	Finish Zone (max 4 points)	- 4 (if autocar stops at the finish zone without exceeding it, including the space above) - 2 (if autocar fully enters finish zone, but the requirement above failed to do so)
E	2		

Stage 2: “Robo Volley Rumble”

Main Task: Contestants are required to control their autocar **manually** to **push** the **ping pong balls**, which spawn periodically in their **own team area**, to the **opponent area**, to achieve the winning condition.

6.1 Additional requirements

- a) Contestant's autocar **can move manually** (before the game there will be autocar testing).

6.2 Before the game starts:

- a) 2 contestants have **30 seconds** to set up their robots in their respective area.
- b) After set up time ends, autocars should be **inside** their respective start zones.

6.3 During the game:

- a) Game length: **3 minutes**.
- b) All autocars are **freely moving** in their **own area**. However, it's a **violation** if any **part of the autocar touches** the **opponent area**, or **goes out from the game field**.
- c) **Ping pong balls** will **enter** from ping pong ball hole **periodically** since the game starts:
 - 3 ball (0th second)
 - 2 ball (20th second)
 - 1 ball (40th second)
 - 2 ball (50th second)
 - 2 ball (60th second)
- d) There will be **totally 20 ping pong balls** in the game field **after 1 minute**.
- e) **All** ping pong balls are **allowed** to be **touched and pushed** by any autocars, **as long as** the balls are **in their own area**.
- f) If any ping pong ball falls out from the game field, the ball will be put back to the side which the contestant causes that incident.

6.4 Deciding the winner

- a) The contestant which **does not have ping pong balls at their own area** will be **declared as the winner** (without any balls fall outside the game field by that contestant at that time)
- b) The contestants with **fewer** ping pong balls at his/her area after **the game ends** will be the winner.
- c) If there's a tie, **1 minute additional time** will be added.
- d) If there's still a tie, each contestant will get 1.5 marks.

6.5 Points

- a) Win - 3 points

7. Retry

- 7.1 To ask for retry, contestants **MUST** raise his/her hand and **SHOUT RETRY**.
- 7.2 While retrying, contestants are allowed to touch and configure his/her autocar, but **cannot change the source code** in blue pill of the autocar.
- 7.3 Contestants may **retry at any checkpoints** which had earned points before. **Retry at the start zone** if there are **no checkpoint** to choose.
- 7.4 There is **no limit for retries**.

8. Violation

- a) Contestants may get a violation if:
 - Disobey or **violate the rules** stated above.
 - False start.
 - **Retry without permission**.
- b) Judge will shout VIOLATION to the contestant, the one who got the violation.
- c) The contestants will be **FORCED TO RETRY** if they get a violation.
- d) **No marks** will be **deducted**. However, getting **violations too frequent** might **lead to DISQUALIFICATION** of the contestant.

9. Disqualification

9.1 Contestant/team will be disqualified if:

- Display the action which violates the spirit of gameplay.
- Intentionally destroy the game field or opponent's autocar.
- Failed to obey the instructions or warnings given by judges.
- Absent in any matches during the autocar day.

10. Contest flow

a) Preliminary Round

- Every contestant will have chances to play stage 2 and stage 1.
- List of the contestants promoted to the playoff section will be arranged based on total points earned.

b) Playoff

- Contestants will play stage 2 followed by stage 1.
- The winner may advance to the next round.

11. Prizes

a) Game:

- Champion
- 1st runner up
- 2nd runner up

b) Special award:

- Best design award

12. Others

12.1 The legitimacy of any actions not mentioned in this rulebook is subject to the discretion of the judges.

12.2 Judges have the **final word** of any judgement during the game.

13. FAQ

13.1 Any questions about game rules may ask **any mentors**, or through this link:

<https://forms.gle/Xk28NBB3TzZbp3LX7>

13.2 Answers will be published weekly on discord.

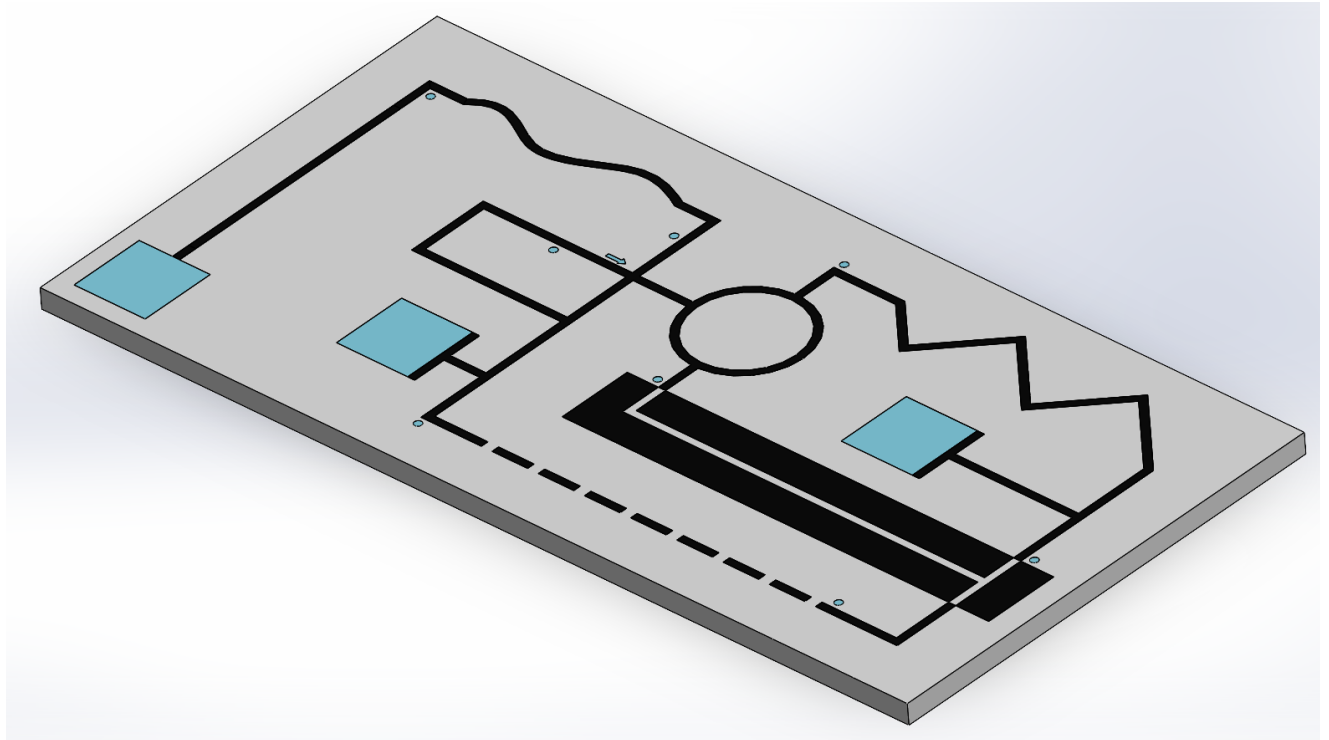
13.3 Participants are **VERY** encouraged to ask questions. However, participants should read through all the game rules before asking.

Unlegit and misunderstanding questions will not be answered.

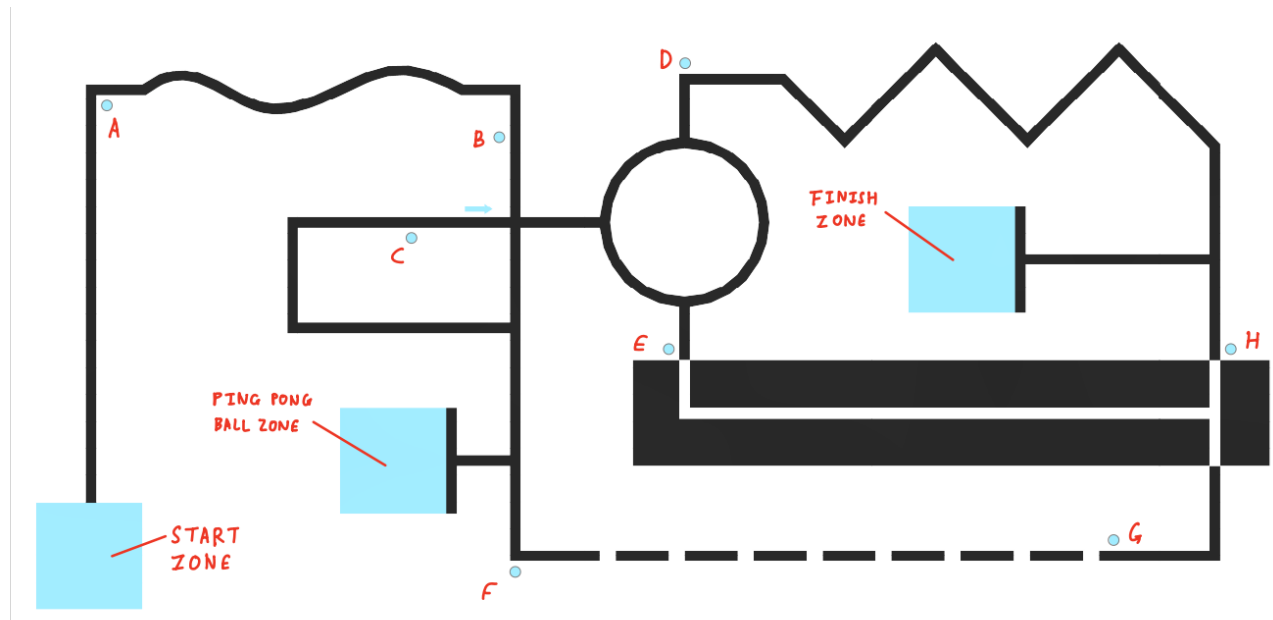
14. Appendix

14.1 Game field 1

a) Perspective view



b) Top view



14.2 Game field 2

a) Perspective view

